

Forecasting Ontario Monthly Energy Demands using SARIMA Models

STATS 4A03

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Section 1: Introduction

Predicting energy needs has become increasingly complex as energy infrastructure improves. We aim to test if a SARIMA model can reliably provide these month-ahead forecasts needed to help grid operators in managing immediate monthly supplies and maintaining stable pricing. Additionally, longer term forecasts such as 1-to-2 year ahead forecasts can support in scheduling power plant maintenance or help plan long-term infrastructure as energy demands increase (Uti).

The following data consists of hourly Ontario electricity demand obtained from the High-Frequency Electricity Data (HEFD) visualization tool provided by the Canadian Center for Energy Information (CCEI). The HEFD tool compiles electricity system data from publicly available sources across Canada and presents them for analysis and research purposes (HFED, 2025).

The Ontario dataset reported through the HEFD tools is collected from the Independent Electricity System Operator (IESO) data directory. The dataset spans the period from January 1, 2014 to April 5, 2026. The primary variables of interests are electricity demand, measured in megawatts (MW) and the corresponding date and time of each observation. Although the original dataset is recorded hourly, the data was aggregated to a monthly level to allow for analysis on broader trends and seasonal patterns in electricity demand.

Section 2: Modelling

The SARIMA model is widely used statistical method for modeling and forecasting time series data which exhibit trend and seasonal patterns which makes it suitable for monthly electricity data which typically display strong seasonal variation due to weather patterns, . The model is denoted $SARIMA(p, d, q) \times (P, D, Q)_s$ where (p, d, q) denote the non-seasonal components, (P, D, Q) denote the seasonal components and s denotes the seasonal period. The model was constructed using the Box-Jenkins methodology.

The autocorrelation function measures the measure the correlation between the current value and past values by a given lag. Gradual decay patterns suggests non-stationarity or trend while sharp cutoffs after certain lags suggest a moving average component (MA). Significant correlations at specific lags (e.g. 12 for monthly data) suggests seasonal moving average components.

The partial autocorrelation function measures correlation between observations seperated by a given lag after removing the effects of intervening variables. The PACF is commonly used to identify autoregressive (AR) components with significant early spikes in the PACF at early

lags suggesting the presence of an autoregressive component. Recurring spikes at specific intervals suggests seasonal autoregressive components (Cryer and Chan, 2008).

The performance of the SARIMA model was evaluated using rolling origin cross validation with an expanding training set generating 1 to 12 month ahead forecasts. Forecasts errors from each iteration were used to compute MAE and MSE.

$$RMSE : \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}$$

$$MAE : \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i|$$

Figure 1 displays the Ontario energy demand from 2014-2026 revealing a slight upward trend from years 2020-2026 revealing possible non-stationarity. Figure 2 displays a boxplot of the energy demands by month revealing a clear seasonal cycle, peaking at ~17,000MW (Jan, Feb, and Jul, Aug) due to increased energy demand from heating and cooling.

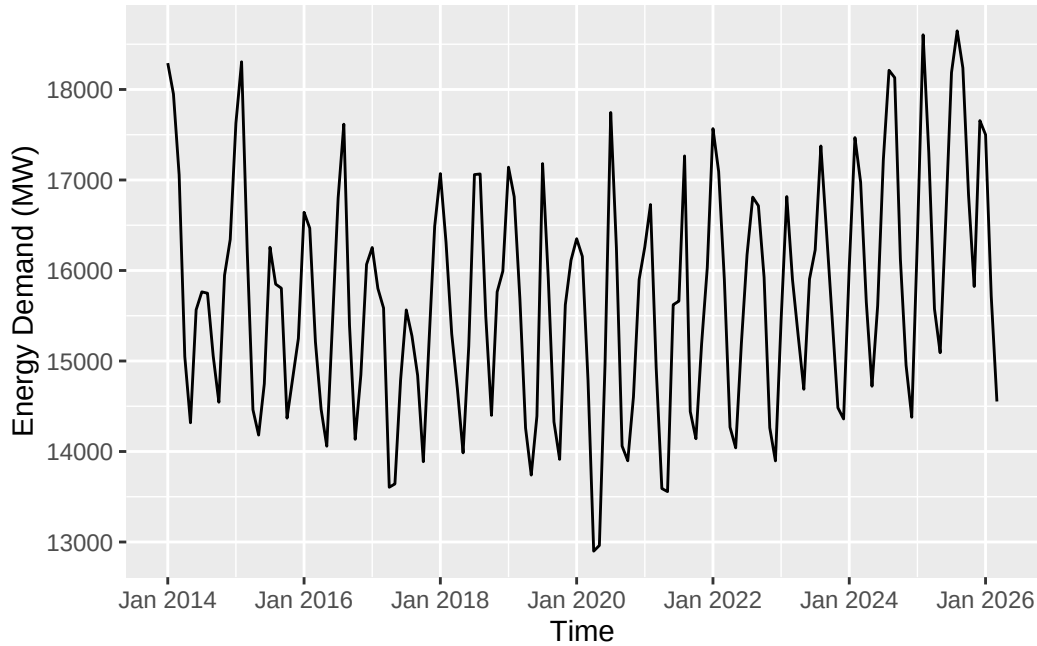


Figure 1: Ontario Energy Demand 2014-2026

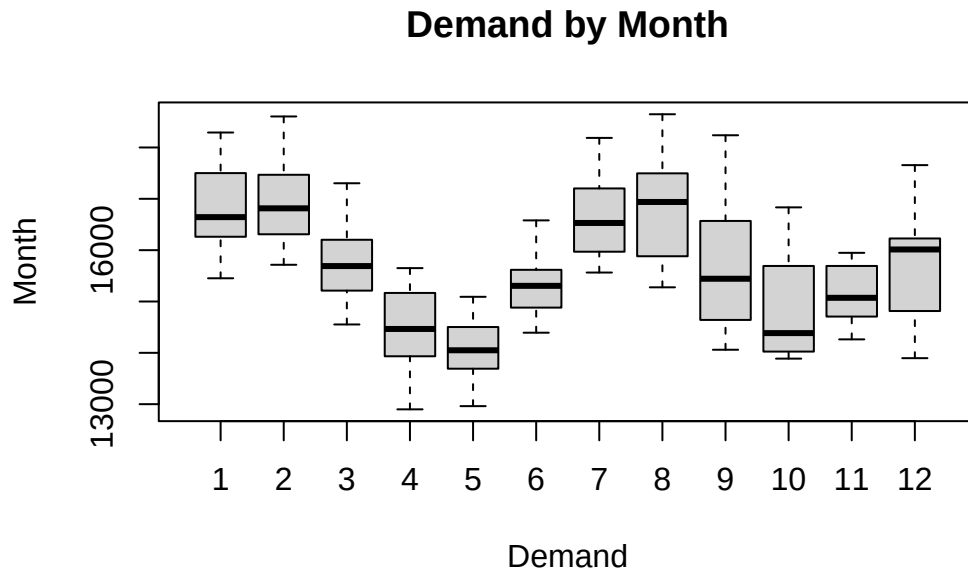


Figure 2: Boxplot of Monthly Energy Demand

Figure 3 displays the ACF and PACF plots for the ontario monthly demand. The ACF plot shows clear seasonal pattern but no apparent gradual decay. The PACF plot shows spikes at lags 1 and 2. Seasonal differencing is needed.

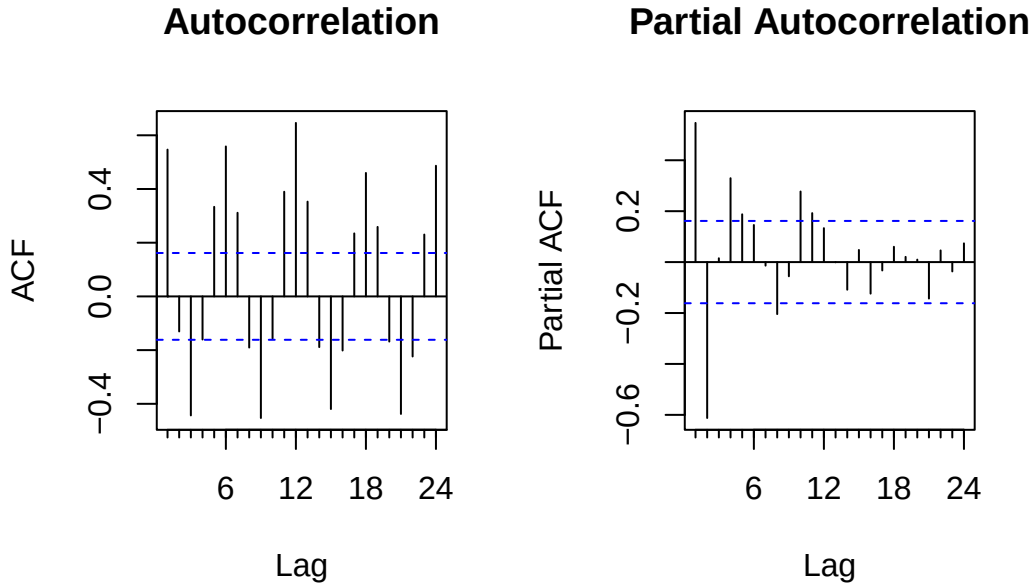


Figure 3: ACF and PACF plots of monthly energy demand

Two candidate models were generated. Figure 4 displays the ACF and PACF plots for the seasonal difference ($D=1$). The ACF plot shows no apparent seasonality with spikes at lags 1, and 12. The PACF plot also shows spikes at lags 1, and 12. A tentative $SARIMA(1, 0, 1) \times (1, 1, 1)_{12}$ model was Due to the slight trend the first difference was also taken along with the seasonal difference ($d=1, D=1$). Figure 5 shows displays the ACF and PACF plots for the dataset with both differences. The ACF plot also shows no seasonality with spikes at lags 1, and 12. The PACF plot shows spikes at lags 1, 3, and 12. A tentative $SARIMA(3, 1, 1) \times (1, 1, 1)_{12}$ model was specified.

Figure 6 displays the diagnostic plots and summary table for the $SARIMA(1, 0, 1) \times (1, 1, 1)_{12}$ model and Figure 7 display the diagnostic plots for the $SARIMA(3, 1, 1) \times (1, 1, 1)_{12}$ model. The first candidate models histogram (top) appears to be slightly left skewed with one significant spike in the ACF residual plot (bottom left) near lag 12. The second candidate models histogram also appears to be left skewed but no significant spikes appear on the ACF residual plot. Both models appear to oscillate around 0. The Ljung-Box test suggests no significant autocorrelation in both models ($p = 0.2491, 0.3084$) implying good fit. The second candidate model appears to show equivalent or better diagnostics and metrics overall and was chosen as the final model.

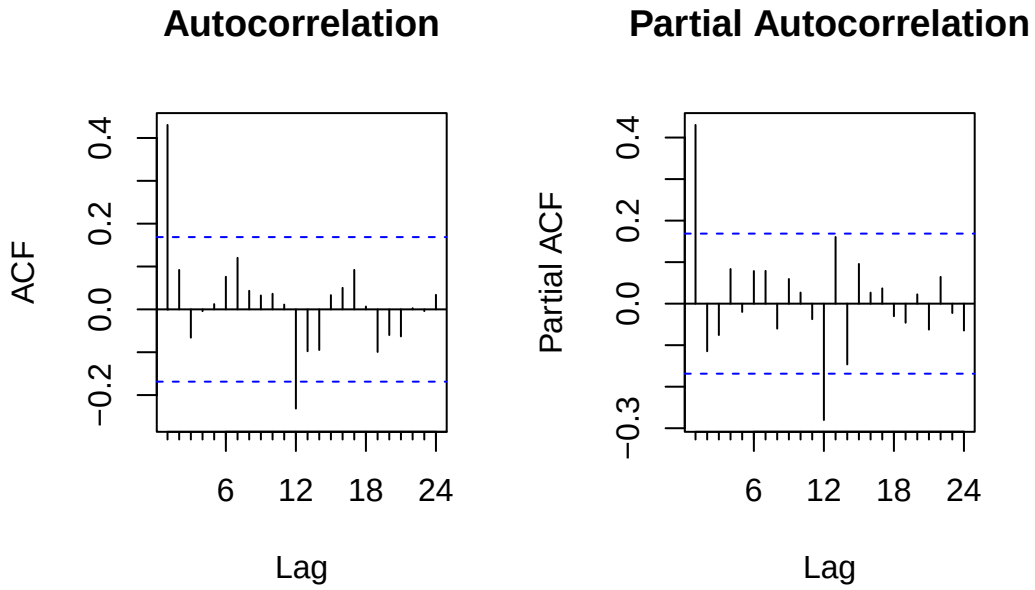


Figure 4: ACF and PACF plots of seasonal difference monthly energy demand ($D=1$)

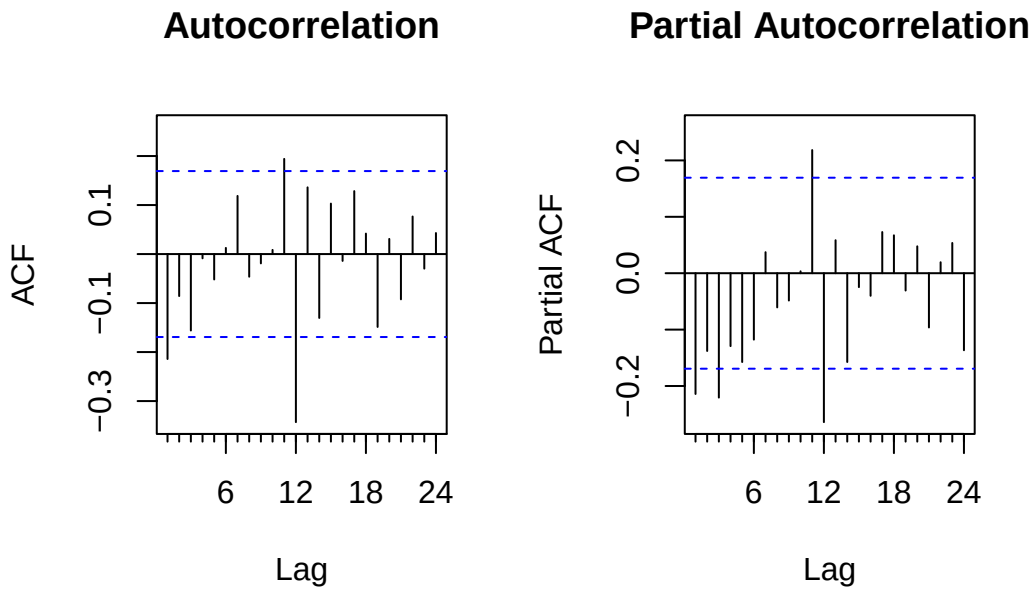


Figure 5: ACF and PACF plots of difference and seasonal difference energy demand ($d=1, D=1$)

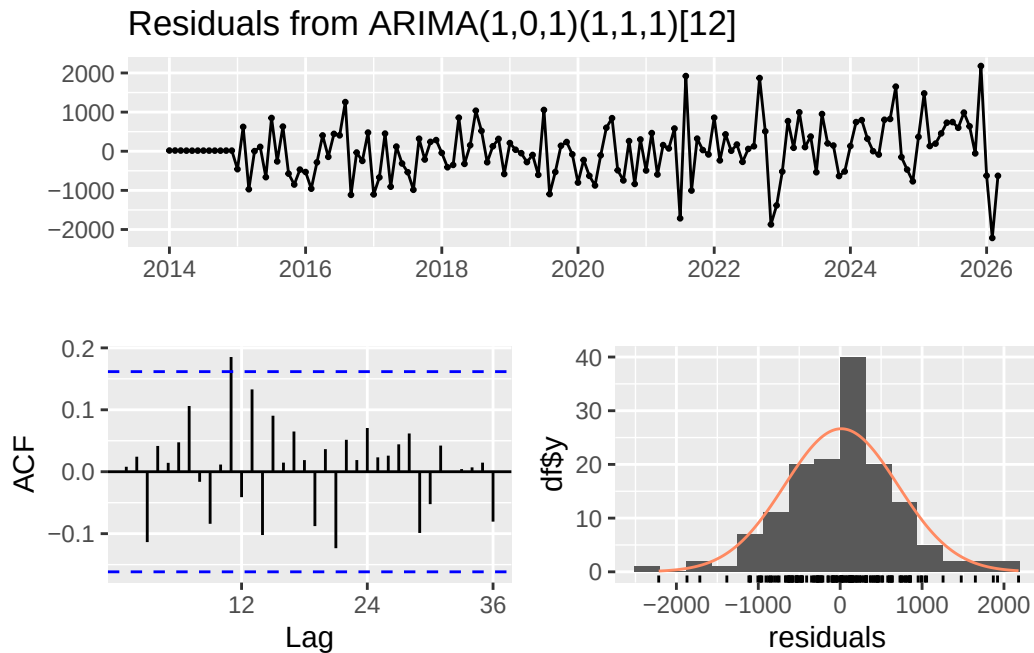


Figure 6: Summary of SARIMA(1,0,1)(1,1,1)[12] model

Ljung-Box test

data: Residuals from ARIMA(1,0,1)(1,1,1)[12]
 Q* = 23.848, df = 20, p-value = 0.2491

Model df: 4. Total lags used: 24

Series: month.sub
 ARIMA(1,0,1)(1,1,1)[12]

Coefficients:

	ar1	ma1	sar1	sma1
	0.4583	0.1750	0.2991	-0.8935
s.e.	0.1274	0.1328	0.1611	0.2068

sigma² = 532292: log likelihood = -1085.83
 AIC=2181.67 AICc=2182.13 BIC=2196.19

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE
Training set	13.45705	688.7349	510.6142	-0.07269122	3.212838	0.7146511

ACF1

Training set	0.00803083
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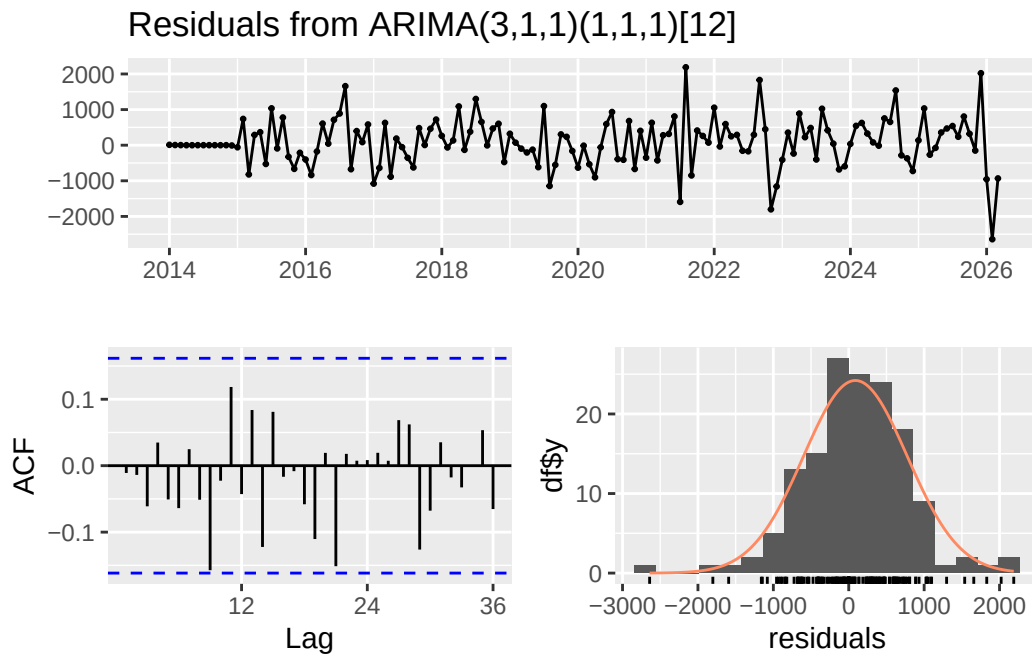


Figure 7: Diagnostic of SARIMA(3,1,1)(1,1,1)[12] model.

Ljung-Box test

data: Residuals from ARIMA(3,1,1)(1,1,1)[12]
 Q* = 20.444, df = 18, p-value = 0.3084

Model df: 6. Total lags used: 24

Series: month.sub
 ARIMA(3,1,1)(1,1,1)[12]

Coefficients:

	ar1	ar2	ar3	ma1	sar1	sma1
	0.5367	-0.1947	-0.1247	-0.9002	0.1287	-0.7520
s.e.	0.1005	0.1089	0.0997	0.0557	0.1670	0.1436


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sigma^2 = 549071:  log likelihood = -1077.39
AIC=2168.78  AICc=2169.67  BIC=2189.06
```

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	87.73608	691.4494	508.5217	0.4318028	3.21018	0.7117225	-0.01095849

Figure 8 displays the RMSE and MAE at various forecast horizons using rolling origin cross validation. Errors appear to sharply increase after each period with significant spikes after horizons $h=1$, $h=12$. Figure 10 shows the 2-year forecast from the final model as well as the fitted vs observed plot. We see in the fitted vs observed plot that our model does appear to underestimate during peak seasons.

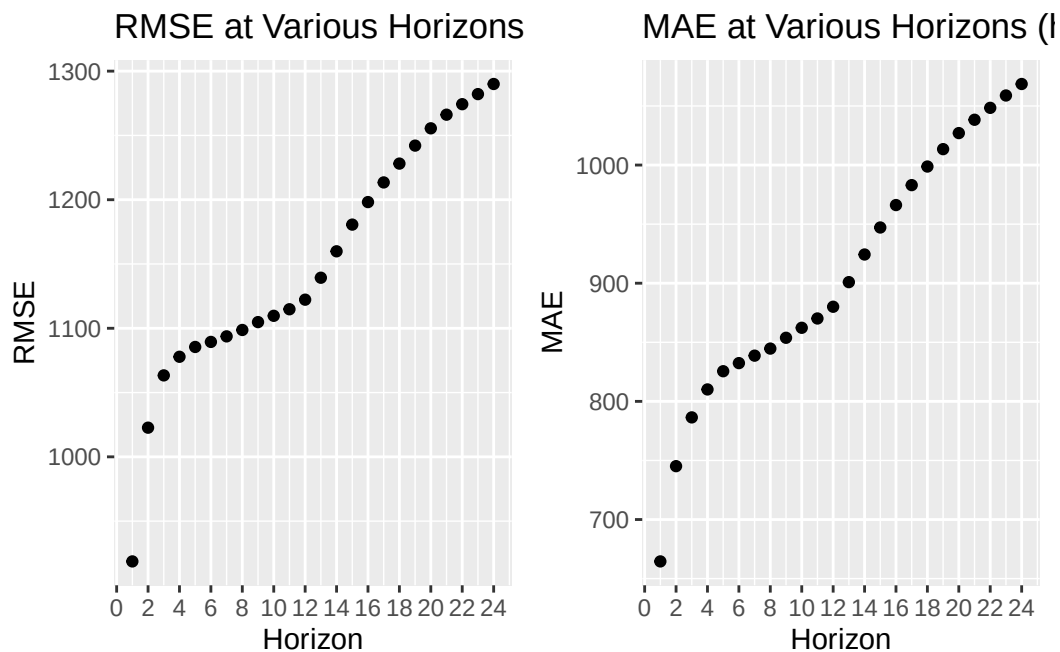


Figure 8: RMSE and MAE of SARIMA(3,1,1)(1,1,1)[12] using CV

SARIMA(3,1,1)(1,1,1)[12]: Observed vs Fitted

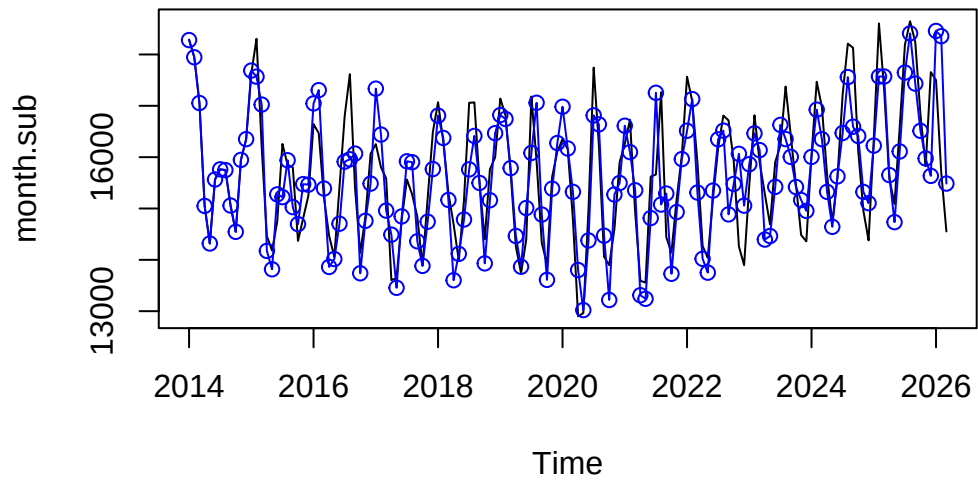


Figure 9: Fitted vs Observed

2-Year Forecast from SARIMA Model

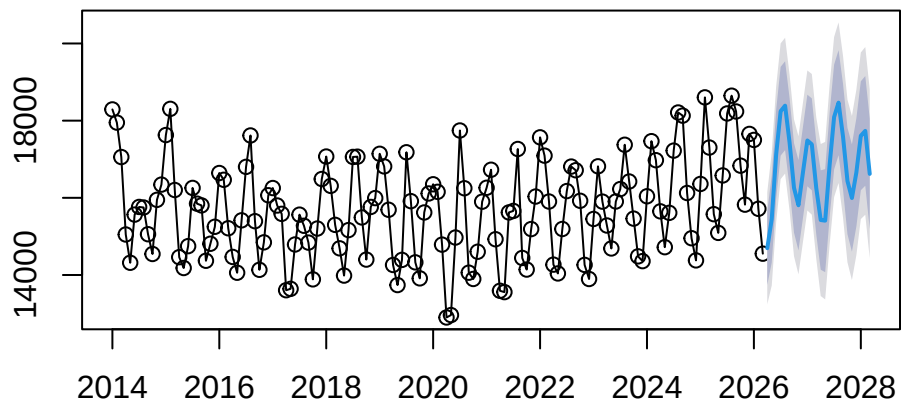


Figure 10: 2-Year Forecast

Section 3: Results

Using a dataset of 98,550 hourly records, aggregated to a monthly level, the SARIMA model effectively captured both linear and seasonal trends in Ontario's monthly demand. Forecast accuracy was evaluated using cross validation across a range of horizons with RMSE and MAE as metrics. The model appears to show strong short term modeling with one month ahead forecasts producing the the lowest error metrics with an RMSE of 918.67 and MAE of 664.58. Longer term forecasts showed increased forecast errors with notable spikes after the 1 and 12 horizons. The 12 and 24 month forecasts still performed well with RMSE of 1122.24, 1289.99 and MAE of 880.08, 1068.61 respectively. Overall the SARIMA model showed strong predictive performance across all metrics. These results seem to suggests that the SARIMA model can provide meaningful support when

Section 4: Conclusion

This study examined the use of SARIMA models to forecast monthly energy demand using historical observations from January 2014 to March 2026 reported from the IESO. The model demonstrated strong performance in capturing seasonal and overall trends in energy demand with RMSE of 918.67 and MAE of 664.58 in monthly forecasts and RMSE of 1122.24, 1289.99 and MAE of 880.08, 1068.61 for 1-to-2 year ahead forecasts.

Despite its effectiveness, the SARIMA model fails to include external factors which may affect energy consumption, such as temperature, policy changes, or economic activity. As a result the model may fail to respond to sudden changes or unusual patterns that are not reflected in the past observations. The model also appears to be extremely accurate when forecasting energy demands between seasons but underestimates peak seasons. Models such as SARIMAX which include an exogenous variable or machine learning methods may be a possible solution. Exploring different levels of aggregation such as daily, or weekly also may also reveal patterns not observed in the monthly aggregation.

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